

Skills Summary

Dividing Complex Numbers with Conjugates

Use this reference while completing your worksheet or when studying.

A quotient is not finished until the denominator is real. The conjugate of $a + bi$ is $a - bi$: same real part, opposite imaginary part. Multiplying top and bottom by it never changes the value, only the form.

1. Why the conjugate clears a denominator

Rule: $(a + bi)(a - bi) = a^2 - b^2i^2 = a^2 + b^2$, a real number. The outer and inner products cancel, and the $-b^2i^2$ becomes $+b^2$. So the conjugate is the one multiplier guaranteed to make a denominator real.

Example: Multiply $6 + 5i$ by its conjugate.

Step 1. The conjugate flips the sign of the imaginary part only, so it is $6 - 5i$; the point is to watch the i terms disappear.

Step 2. $(6 + 5i)(6 - 5i) = 36 - 30i + 30i - 25i^2 = 36 + 25$.

61

Try it: Multiply $3 + 7i$ by its conjugate.

check: 58

2. Dividing by a pure imaginary number

Rule: the conjugate of bi is $-bi$, and $(bi)(-bi) = b^2$. Useful shortcut: $\frac{1}{i} = -i$, so dividing by i is multiplying by $-i$. Write the answer as $a + bi$, real part first.

Example: Simplify $\frac{10 + 15i}{5i}$.

Step 1. The denominator has no real part, so its conjugate is just $-5i$ — one multiplication clears it.

Step 2. $\frac{(10 + 15i)(-5i)}{(5i)(-5i)} = \frac{-50i + 75}{25} = \frac{75 - 50i}{25}$.

3 - 2i

Try it: Simplify $\frac{8 - 2i}{2i}$.

check: -1 - 4i

3. Dividing by a full complex denominator

Rule: multiply top and bottom by the conjugate of the denominator, then divide *both* parts of the new numerator by the real denominator: $\frac{z}{a+bi} = \frac{z(a-bi)}{a^2+b^2}$. Dividing real by real and imaginary by imaginary is not a legal move.

Example: Simplify $\frac{11+3i}{3+2i}$.

Step 1. The denominator is a full binomial, so the conjugate $3-2i$ is what turns the bottom into the real number $9+4$.

Step 2. $\frac{(11+3i)(3-2i)}{3-i} = \frac{33-22i+9i+6}{13} = \frac{39-13i}{13}$.

Try it: Simplify $\frac{5+5i}{1+3i}$.

check: 2 - 2i

4. Circuits: voltage, current and impedance

Model: in an AC circuit $V = I \cdot Z$, where voltage V is in volts, current I in amps and impedance Z in ohms — all three complex. So $Z = V/I$ and $I = V/Z$: every missing quantity is one division by the quantity you already know.

Example: $V = 1 + 7i$ volts across an impedance of $1 + 2i$ ohms. Find the current.

Step 1. Current is wanted and V and Z are known, so rearrange $V = IZ$ to $I = V/Z$ — a division by the impedance.

Step 2. $\frac{(1+7i)(1-2i)}{1+4} = \frac{1-2i+7i+14}{5} = \frac{15+5i}{5}$.

Try it: $V = 4 + 7i$ volts, $Z = 2 + i$ ohms. Find the current in amps.

check: 3 + 2i